Summary in Graph

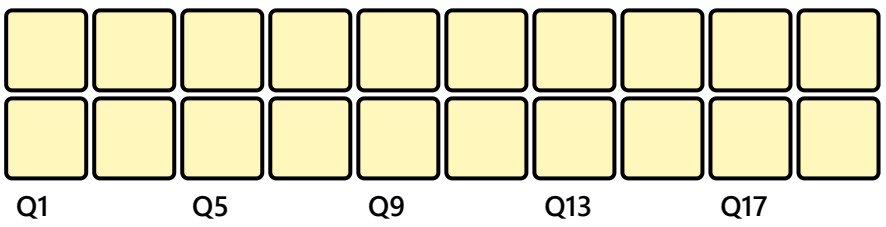
Exam Summary (GO Classes CS/DA | Probability Distributions | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	20 (10 + 10)
Total Marks:	30 (10 + 20)
Exam Duration:	60 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (52) out of 521 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Probability

Suppose Alice rolls a 6-sided die, and Bob rolls a 4-sided die. Let X denote the maximum value on the two dice. Find $P(X = 6)$.

A. $1/24$
B. $3/24$
C. $7/24$

D. $4/24$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Consider two independent Binomial random variables X and Y with parameters $n = 5$ and $p = 1/2$. Determine which of the following options correctly describes the random variable represented by $\mathrm{X} + \mathrm{Y}$.

A. $\mathrm{X} + \mathrm{Y}$ is a Binomial random variable with parameters $n = 5$ and $p = 1/2$.
B. $\mathrm{X} + \mathrm{Y}$ is a Binomial random variable with parameters $n = 10$ and $p = 1/2$.
C. $\mathrm{X} + \mathrm{Y}$ is a Binomial random variable with parameters $n = 5$ and $p = 1/4$.
D. $\mathrm{X} + \mathrm{Y}$ is a Binomial random variable with parameters $n = 10$ and $p = 1/4$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose that X, Y, and Z are independent random variables. $\mathbf{E}[X] = 1, \mathbf{E}[Y] = 2$, and $\mathbf{E}[Z] = 3$. $\mathrm{var}(X) = 1$, $\mathrm{var}(Y) = 2$, and $\mathrm{var}(Z) = 3$. Compute the value of the following expression:

$$\mathbf{E}[2X + 3Y - Z - 10] + \mathrm{var}(Y - 2X + 2Z)$$

A. 5
B. 13
C. -3
D. 10

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Assuming that X and Y are non-negative random variables, which may or may not be independent. Which of the following statements is/are ALWAYS true?

A. $P(X + Y = 2) \leq P(X = 1)P(Y = 1)$
B. $P(X + Y = 2) \geq P(X = 1)P(Y = 1)$
C. $P(X + Y = 2) \leq P(\{X \geq 1\} \cup \{Y \geq 1\})$
D. $P(X + Y = 2) \geq P(\{X \geq 1\} \cup \{Y \geq 1\})$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a discrete uniform random variable. It has a probability mass function (PMF) $P(X = k) = \dfrac{1}{n}$ for $k = 1, \ldots, n$, the expected value $\mathbb{E}[X]$ is given by

A. $\frac{n(n+1)}{2}$,
B. $\frac{n^2}{2}$

C. $\frac{n+1}{2}$
D. $\frac{n}{2}$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

During a lecture, an instructor picks a student at random and asks if they know a certain definition from the course. Define the following two random variables X and Y. X takes the value 0 or 1 where $X = 1$ if the student knows the answer and $X = 0$ otherwise. Y is the number of the student's discussion section, i.e., Y is either $1, 2,$ or 3. The joint probabilities are given in the following table:

	$Y = 1$	$Y = 2$	$Y = 3$
$X = 0$	0.2	0.2	0.2
$X = 1$	0.05	0.05	0.3

So, for example $P(X = 0, Y = 1) = 0.2$ and $P(X = 1, Y = 1) = 0.05$. What is the value of $P(X = 0 \mid Y = 1)$?

A. 0.2
B. 0.8
C. 0.6
D. 0.4

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Suppose that X and Y are independent random variables with $\mathrm{Var}(X) = 1, \mathrm{Var}(Y) = 2$. Find $\mathrm{Var}(1 - 2X + 3Y)$.

Your Answer: | Correct Answer: 22 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be Bernoulli random variables with parameter $p(\text{i.e.}, P(X = 1) = p, P(X = 0) = 1 - p)$.

Define a random variable $Y = 2^X 3^{1-X}$.

What is $E[Y]$?

A. p
B. 2^p
C. $3 - p$
D. $\log_6 p$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a random variable with PDF

$$f(x) = \begin{cases} C\left(1-x^2\right) & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

What is C?

A. $1/2$
B. $1/3$
C. $1/4$
D. $3/4$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a continuous uniform random variable on $[0, 1]$. Let $a \in [0, 1]$ be a constant, find the probability $P(X > a)$?

A. $(1-a)$
B. $(1-a)/2$
C. $2(1-a)$
D. $(1-a)^2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X_i are independent discrete uniform random variables defined for consecutive integers in the interval $[1, n]$. That is $P(X_i = j) = 1/n$ for any i and j in $[1, n]$.

Let $Y = \min\{X_1, X_2, \ldots, X_n\}$.

Find $P(Y > k)$ for some $k = n - 1$.

A. $(1/n)^{\mathrm{n}}$
B. $(1 - 1/n)^n$
C. $1 - (1/n)^n$
D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose that X and Y are independent random variables,

such that X has probability mass function $P(X = x) = (1/3)(2/3)^{x-1}$ for integers $x \geq 1$,

and Y has probability mass function $P(Y = y) = (2/5)(3/5)^{y-1}$ for integers $y \geq 1$.

Find $P(X = Y)$.

A. $1/9$
B. $2/9$
C. $1/3$

D. $1/2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider the scenario where you keep tossing the coin until you get a head. Let p represent the probability of getting a head and q (which equals $1-p$) represent the probability of getting a tail.

Let Y be a random variable that counts the number of tosses until the head appears. What is the probability that Y is a multiple of 3? To calculate this probability assume the coin is fair.

A. $1/3$
B. $1/4$
C. $3/4$
D. $1/7$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Numerical Type | Award: 2 | Penalty: 0 | Probability

Suppose that a person wins a game of chance with probability 0.40, and loses otherwise. If he wins, he earns 5 dollars, and if he loses, then he loses 4 dollars. Assume that he plays ten games independently. Let X denote the number of games that he wins. What is his expected gain or loss (altogether) during the ten games?

[Hint: X is a binomial random variable with k successes (or wins) but It is not asked to find $E[X]$, you first need to find gain/loss in terms of X then need to find the expectation of that term]

Your Answer: | Correct Answer: -4,4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

You take the bus, walk, or ride your bike to school. In the morning you toss a fair, six-sided die.

- If you get a 1 or 2, you take the bus.
- If you get a $3, 4$, or 5, you walk.
- If you get a 6, you ride your bike.

On average, it takes you 12 minutes if you take the bus, 14 minutes if you walk, and 6 minutes if you ride your bike. What is the expected amount of time it takes you to get to campus?

A. 9 minutes
B. 10 minutes
C. 11 minutes
D. 12 minutes

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A random variable X has a probability mass function given by

$$p_X(x) = \begin{cases} \frac{1}{4}, & \text{if } x = 1 \\ \frac{1}{2}, & \text{if } x = \frac{3}{2} \\ \frac{1}{4}, & \text{if } x = 3 \\ 0, & \text{otherwise.} \end{cases}$$

Find the cumulative distribution function, $F_X(x)$, of X. Compute the value of the following expression:

$$\mathbf{E}[X] + F_X(2)$$

A. $3/4$
B. $7/4$
C. 2
D. $5/2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A die is rolled until two different numbers appear. Let T be the total number of times the die is rolled. Obviously $\mathrm{P}(\mathrm{T}=0)=\mathrm{P}(\mathrm{T}=1)=0$.

For example,

$\{1,1,2\}, \mathrm{T}=3$
$\{1,0\}, \mathrm{T}=2$
$\{0,0,0,0,5\}, \mathrm{T}=5$
$\{4,4,4,4,4,6\}, \mathrm{T}=6$

Which of the following is the CORRECT PMF for T?

A. $\mathrm{P}(\mathrm{T}=k)=\left(\frac{5}{6}\right)^{k-2} \cdot \frac{1}{6}, \quad \text{For } k \geq 2$
B. $\mathrm{P}(\mathrm{T}=k)=\left(\frac{1}{6}\right)^{k-2} \cdot \frac{5}{6}, \quad \text{For } k \geq 2$
C. $\mathrm{P}(\mathrm{T}=k)=\left(\frac{1}{6}\right)^{k}, \text{ For } k \geq 2$
D. $\mathrm{P}(\mathrm{T}=k)=\left(\frac{5}{6}\right)^{k}, \text{ For } k \geq 2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $\mathrm{X}_1, \mathrm{X}_2$ be independent. Bernoulli random variables with parameter p (i.e., they are independent and satisfy $P(X_i=1)=p, P(X_i=0)=1-p$) Find $E\left[X_1^2 X_2\right]$?

A. p^3
B. p^2
C. p
D. $1-p^2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Multiple Choice Type Award: 2 Penalty: 0.67 Probability

A game is played with a weighted coin that is heads with probability 0.6 and tails with probability 0.4. The game will consist of a number of rounds. In each round, you flip the coin twice, and.

- if you get HH, you win $\$5$ and you move on to the next round.
- if you get HT or TH, you win $\$1$ and you move on to the next round.
- if you get TT, the game is over.

Compute your expected winnings from playing this game.

A. $80/3$
B. $57/4$
C. 25
D. $32/3$

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 07sec Discuss

Q #20 Multiple Choice Type Award: 2 Penalty: 0.67 Probability

Suppose that X, Y and Z are independent random variables such that takes values either 0 or 1 with equal probability 0.5. What will be the $P[X+Y+Z=1 \mid X=Y]=?$

A. $1/2$
B. $1/4$
C. $1/8$
D. $1/16$

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

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